



Information Technology
Generalist

PROG1400 – MidTerm Exam: Object-Oriented Programming Concepts

Course: PROG1400 – Introduction to Object-Oriented Programming

Instructor: Davis Boudreau

Learning Outcome Assessed:

- LO1 – Describe applications using core principles of object-oriented programming

Exam Details

- **Case Study:** Pac-Man
- **Student Context:** Your Pac-Man-like game project
- **Time:** 90 minutes
- **Total Marks:** 50
- **Exam Type:** Closed Book
- **Computer Use:** Not Permitted

Instructions to Students

- This is a **closed-book exam**
- **No computers, phones, or electronic devices** are permitted
- Answer **all questions**
- Use **clear explanations** and **correct object-oriented terminology**
- Diagrams are optional but must be **hand-drawn**
- **Do not write code** — this exam assesses your ability to **describe** object-oriented design

Question 1 – Pac-Man as an Object-Oriented Application (10 marks)

- a) Explain why **Pac-Man** is well suited to being designed as an object-oriented application. (5 marks)
- b) Identify **two objects** from the Pac-Man game and describe the **responsibility** of each object. (5 marks)
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Question 2 – Your Game as an Object-Oriented System (10 marks)

- a) Briefly describe your game project and its main objective. (3 marks)
- b) Identify **three objects** in your game and describe **what each object is responsible for**. (7 marks)

Question 3 – Encapsulation (8 marks)

- a) Describe what **encapsulation** means in the context of an object-oriented game application. (4 marks)
- b) Using **one object from Pac-Man** and **one object from your own game**, explain:
- What data the object should control
 - Why other objects should not directly access or modify that data (4 marks)

Question 4 – Abstraction (8 marks)

- a) Explain **abstraction** using **Pac-Man and Ghosts** as examples. (4 marks)
- b) In your own game, identify **two objects** that share similar behavior and explain how abstraction helps describe them at a higher level. (4 marks)

Question 5 – Inheritance and Polymorphism (8 marks)

- a) Explain how **inheritance** could be used in Pac-Man or your game to reduce duplicated behavior. (4 marks)
- b) Describe an example of **polymorphism** that could exist in Pac-Man or your game. (4 marks)

Question 6 – Aggregation and Interfaces (6 marks)

- a) Explain **aggregation** in a game application using Pac-Man or your game as an example. (3 marks)
- b) Describe the purpose of an **interface** in an object-oriented game design and give one example related to movement, collision, or interaction. (3 marks)

Here is a **cleaned, streamlined version** of your rubric with **duplicates removed**, consistent formatting, and clearer layout.

(You didn't include code—just rubric text—so I cleaned this document.)

Rubric – LO1 Exam

Total: 50 points

Criteria	Exceeds Expectations	Meets Expectations	Approaching Expectations	Below Expectations	Points
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Criteria	Exceeds Expectations	Meets Expectations	Approaching Expectations	Below Expectations	Points
Understanding of OOP Principles	Provides clear, accurate explanations of encapsulation, abstraction, inheritance, polymorphism, aggregation, and interfaces using correct terminology	Mostly accurate explanations with only minor gaps	Partial understanding; explanations are vague or incomplete	Major misunderstandings or incorrect terminology	/20
Application to Pac-Man Case Study	Uses Pac-Man accurately and effectively to explain object responsibilities and OOP concepts	Uses Pac-Man with mostly correct examples; minor inaccuracies	References Pac-Man but explanations lack clarity	Little or incorrect reference to Pac-Man	/15
Application to Student Game Project	Clearly applies OOP concepts to the student's own game with well-defined objects and responsibilities	Applies OOP concepts with minor issues	Superficial or unclear application to student's game	Minimal or incorrect application to student's game	/15

Total Score: /50